OS Orange 2

SRN: PES2UG24CS311

Name: Neeraj R Rugi

Screenshots

1A.

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B.

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$
```

2A.

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B.

```
All Phase 1 tests passed.
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337
afedb9b46a1af8e11ec4f476 | head -20
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$
```

3A.

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ ./pes init
Initialized empty PES repository in .pes/
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ ./pes add file1.txt file2.txt
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h
```

3B.

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ cat .pes/index    # Human-readable text format
100755 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776672354 6 file1.txt
100755 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776672354 6 file2.txt
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$
```

4A.

```
File  Edit  View  Bookmarks  Plugins  Settings  Help
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: 02bbcb52efbd... Initial commit
Committed: 6912db2dd6de... Add world
Committed: fbda1d6390bd... Add farewell
commit fbda1d6390bddab4e013df17ecc37bfd70d830de84ef84dc481b9cb4b98b4d25
Author: Neeraj R Rugi PES2UG24CS311
Date:   1776677104

    Add farewell

commit 6912db2dd6de00031119805ae09e7a4a5417810837a01f6ce07c84ee787bca58
Author: Neeraj R Rugi PES2UG24CS311
Date:   1776677103

    Add world

commit 02bbcb52efbda7eda6c33d9b5ba80af0c6b248551221fc3eea1c1dd440044b9f
Author: Neeraj R Rugi PES2UG24CS311
Date:   1776677103

    Initial commit

neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$
```

4B.

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/02/bbcb52efbda7eda6c33d9b5ba80af0c6b248551221fc3eea1c1dd440044b9f
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/55/8c1061e21bccbeb23c527d8dcef5c8646cabfb6091aae3ae7ee3320c9ff7b7
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/69/12db2dd6de00031119805ae09e7a4a5417810837a01f6ce07c84ee787bca58
.pes/objects/cc/e49eb2943f54e8135f291862bdf218064409ba05a7ee7d5bec00ddc22addfb
.pes/objects/dd/82340b5e403e2dc18bfca050e6699935849710b2b5d28ab8bbffbb72d32627
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/fb/da1d6390bddab4e013df17ecc37bfd70d830de84ef84dc481b9cb4b98b4d25
.pes/refs/heads/main
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$
```

4C.

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ cat .pes/refs/heads/main
fbda1d6390bddab4e013df17ecc37bfd70d830de84ef84dc481b9cb4b98b4d25
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ cat .pes/HEAD
ref: refs/heads/main
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$
```

Final

```
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$ make test-integration
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
═══ Running integration tests ═══
bash test_sequence.sh
═══ PES-VCS Integration Test ═══

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 34242c7be8f0... Initial commit

Log after first commit:
commit 34242c7be8f04f64a8980090818017e1ce9553f48f4d8abac2b98b79a457d1a0
Author: Neeraj R Rugi PES2UG24CS311
Date:   1776677276

    Initial commit


--- Second Commit ---
Committed: 0c40fbdb1bdd... Update file.txt
```

```
--- Second Commit ---
Committed: 0c40fbdb1bdd... Update file.txt

--- Third Commit ---
Committed: f850a9919b75... Add farewell

--- Full History ---
commit f850a9919b753d608f79e529bbc06e89580a1f7afe0bbf33e6ee519b6b5a962b
Author: Neeraj R Rugi PES2UG24CS311
Date:   1776677276

    Add farewell

commit 0c40fbdb1bdd09d3bf86d17027d2d9ee5e8870b683e253c78b3c4bce9edd561c
Author: Neeraj R Rugi PES2UG24CS311
Date:   1776677276

    Update file.txt

commit 34242c7be8f04f64a8980090818017e1ce9553f48f4d8abac2b98b79a457d1a0
Author: Neeraj R Rugi PES2UG24CS311
Date:   1776677276

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
f850a9919b753d608f79e529bbc06e89580a1f7afe0bbf33e6ee519b6b5a962b

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/0c/40fbdb1bdd09d3bf86d17027d2d9ee5e8870b683e253c78b3c4bce9edd561c
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/34/242c7be8f04f64a8980090818017e1ce9553f48f4d8abac2b98b79a457d1a0
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/f8/50a9919b753d608f79e529bbc06e89580a1f7afe0bbf33e6ee519b6b5a962b

═══ All integration tests completed ═══
neeraj-r-rugi@neeraj-kubuntu:/media/neeraj-r-rugi/NEERAJ-DRIVE/BOJ_Problems/OS_Orange_last$
```

Questions

Q5.1

In PES (and Git), a branch is represented as a file within `.pes/refs/heads/` that holds a commit hash. Implementing `pes checkout <branch>` requires the following updates:

Files modified under `.pes/`:

1. **HEAD file** — `.pes/HEAD` must be updated to reference the target branch: `ref: refs/heads/<branch>`

2. **Object store** — No modifications are necessary. Objects are immutable and remain untouched.

Beyond updating HEAD, the working directory must be reconstructed to match the state of the checked-out branch. This involves reading the commit hash from the branch file, loading its associated tree object, and rebuilding the directory structure by restoring files from blob objects in the object store.

This process is non-trivial: it requires recursive directory reconstruction, correct mapping of object data to file paths, and safe replacement of existing files. Edge cases such as locally modified files must also be handled carefully to avoid unintended data loss.

Q5.2

Before switching branches, the system must verify that the working directory is clean — that is, free of uncommitted modifications. A working directory is considered "dirty" if any tracked file has been altered since its last commit.

Detection works by comparing the working directory against the index (which records the last staged state) and the object store. For each tracked file, metadata such as modification time and size is checked against the values stored in the index. If these differ, the file is flagged as potentially modified. For stronger guarantees, the file's hash can be recomputed and compared with the hash in the index.

The flagged file is then compared against its version in the target branch's tree. If the file is both locally modified and differs in the target branch, a genuine conflict exists — checkout is aborted to protect the user's uncommitted work.

Q5.3

A detached HEAD state arises when `.pes/HEAD` contains a raw commit hash rather than a symbolic reference to a branch. In this state, the user is not positioned on any named branch.

Any commit made in detached HEAD creates a new commit object in the store, but since no branch reference is updated to point to it, the commit is effectively orphaned. It is not reachable from any branch head, making it a "dangling" commit. While it persists in the object store initially, it becomes eligible for deletion during garbage collection once no references point to it.

Recovery is possible as long as the commit hash is known — for example, from the log. The user can create a new branch pointing to that hash, which reattaches the commit to the repository's reachable history.

Q6.1

As a repository evolves, the object store accumulates unreachable objects — commits, trees, and blobs that are no longer referenced by any branch. Garbage collection reclaims this space through a reachability traversal.

The process begins from all branch heads in `.pes/refs/heads/`. Starting from each branch's tip commit, the traversal follows parent commit links, associated tree objects, and their blob entries, collecting every encountered hash into a reachable set. A hash set data structure is used for O(1) membership checks during this marking phase.

Once marking is complete, every object in `.pes/objects/` is enumerated. Any object whose hash is absent from the reachable set is safe to delete.

For a repository with approximately 100,000 commits across 50 branches, many commits will be shared among branches. In practice, traversal would cover around 100,000 unique commits along with their trees and blobs, resulting in on the order of several hundred thousand objects being examined.

Q6.2

Concurrent execution of garbage collection and a commit operation poses a serious risk of data corruption.

Consider the sequence: a commit begins by writing blob and tree objects to the store, but has not yet updated the branch reference or HEAD. At this intermediate point, the new objects exist in the store but are not reachable from any branch head.

If GC runs concurrently, it scans from branch heads and — finding no reference to the new objects — incorrectly classifies them as unreachable and deletes them. When the commit operation subsequently updates HEAD, it will point to objects that have already been removed, leaving the repository in a corrupt state.

Git addresses this through a combination of atomic file operations and locking. Objects are written using temporary files and renamed atomically into place. GC is coordinated to avoid interfering with ongoing writes, and recently created objects are granted a grace period before they become eligible for collection.